



DELHI PUBLIC SCHOOL NEWTOWN
SESSION 2025-26
MONDAY TEST

CLASS : X
SUBJECT: MATHEMATICS

FULL MARKS: 40
DATE: 23/06/25

General Instructions:

- The paper consists of three printed pages.
- All questions are compulsory.
- Copy the question number carefully before answering the questions.

SECTION: A

1. Choose the correct answers to the questions from the given options. [1 × 8 = 8]
(Do not copy the questions, write the correct answers only)

i) In solving a quadratic equation, one of the values of the variable 'x' is 633.356. The solution rounded to two significant figures is:

- (a) 633.36 (b) 633.35 (c) 633.3 (d) 630

ii) The CGST paid by a customer to the shopkeeper for an article which is priced at ₹500 is ₹15. The rate of GST charged is:

- (a) 1.5% (b) 3% (c) 5% (d) 6%

iii) If 'x' satisfies the inequations $2x - 7 < 11$ and $3x + 4 < -5$, then

- (a) $x < 3$ (b) $x > -3$ (c) $x < -3$ (d) $x > 3$

iv) A man bought a Health Insurance Policy for his family and paid ₹ 900 as SGST. Find the total annual premium paid by him for this policy, rate of GST being 18%.

- (a) ₹11800 (b) ₹10000 (c) ₹5000 (d) ₹3600

v) A person deposited ₹ 300 per month in a recurring deposit account for 18 months. The qualifying sum of money for the calculation of interest is:

- (a) ₹3600 (b) ₹5400 (c) ₹51,300 (d) ₹1,63,800

vi) If $x = -2$ is one of the solutions of the quadratic equations $x^2 + 3a - x = 0$, then the value of "a" is :

- (a) -8 (b) -2 (c) $-1/3$ (d) $1/3$

vii) The discriminant of the quadratic equation $bx^2 + ax + c = 0$; $b \neq 0$ is given by

- (a) $b^2 - 4ac$ (b) $\sqrt{b^2 - 4ac}$ (c) $\sqrt{a^2 - 4bc}$ (d) $a^2 - 4bc$

viii) Assertion(A): If $a < b$, $c < 0$, then $a/c < b/c$.

Reason(R): If both sides are divided by the same negative quantity, then the inequality is reversed.

- (a) A is true, R is false. (b) A is false, R is true.
(c) Both A and R are true, and R is the correct reason for A.
(d) Both A and R are true, and R is incorrect reason for A.

SECTION: B

2. A manufacturer gives a discount of 20% to a wholesaler on an Android tablet marked at ₹15,000. The wholesaler then sells it to a retailer at 10% discount on marked price. If a retailer sells it to the customer at a marked price and GST charged is 18% at each stage of intra state transaction, find the:

- a) cost price of the wholesaler including GST.
b) GST paid by the retailer to the State Govt.
c) price the customer pays for the product including GST. [4]

3. A man deposits ₹250 every month in a bank's recurring deposit account for a period of 3 years. He received ₹10,110 at the time of maturity. Find:

- a) the rate of interest
b) how much more interest he will receive if he had deposited ₹50 more per month at the same rate of interest and for the same time. [4]

4.a) Solve for 'x' using quadratic formula: $2x^2 - 5 = 10x$. Give your answer correct to two decimal places.

- b) Find the value of 'k' for which the given equation has real roots: $9x^2 + 3kx + 4 = 0$
[3 + 3 = 6]

5.a) Solve the following inequation, write the solution set and represent it on the number line: $\frac{1}{2}(2x - 1) \leq 2x + \frac{1}{2} \leq 5\frac{1}{2} + x$; $x \in R$

- b) Write the solution set and also find the smallest value of 'x' such that:

$$-3(2 + x) < 2(3x - 1); x \in W \quad [3 + 3 = 6]$$

6. A man deposited ₹600 per month in a recurring deposit account. The bank pays a simple interest of 12% p.a. Calculate the:

- a) number of monthly instalments he deposits to get a maturity amount of ₹11,826?
b) total interest paid by the bank.
c) total amount deposited by him. [4]

7. Mr A covers a distance of 1.5 km between home and school at a speed of 'x' km/hr. Mr B covers the same distance at a speed which is 4km/hr less than Mr A.

- a) find the time taken by Mr A to reach school.
b) find the time taken by Mr B to reach school.

c) If Mr B takes 6 minutes more than Mr A find the value of 'x'.

d) If Mr B takes 6 minutes more than Mr A find the speed of Mr B.

[4]

8. A customer went to a departmental store and bought the following items. The discount and GST rates and the quantity of each item and marked price of each are given below:

Item	Quantity	Rate per item(₹)	Discount	GST
Trouser	1	₹3,500	10%	12%
Pair of shoes(branded)	2	₹1,200	5%	18%
Whole wheat packets	3	₹500	5%	NIL
Bakery products(unpacked)	4	₹200	NIL	NIL

Calculate the:

a) The total amount of GST paid.

b) The total bill amount paid by him.

[4]